School of Mathematics and Statistics, UNSW
MATH3311/MATH5335 Python
Term 2, 2023

Exercises 2: Vector and Matrix Norms

1. Define $\boldsymbol{x} = [-9, 5, 1, -4, 12, -7]^T$ and calculate the following norms,

   (a) $\|\boldsymbol{x}\|_1$ by hand and using Python.

   (b) $\|\boldsymbol{x}\|_2$ by hand and using Python.

   (c) $\|\boldsymbol{x}\|_\infty$ by hand and using Python.

   You may need to import the followings:

```
from numpy import inf, array
from scipy.linalg import norm
```

2. It can be shown that for $\boldsymbol{x} \in \mathbb{R}^n$

$$
\begin{aligned}
\|\boldsymbol{x}\|_2 &\le \|\boldsymbol{x}\|_1 \le \sqrt{n}\|\boldsymbol{x}\|_2, \\
\|\boldsymbol{x}\|_\infty &\le \|\boldsymbol{x}\|_2 \le \sqrt{n}\|\boldsymbol{x}\|_\infty, \\
\|\boldsymbol{x}\|_\infty &\le \|\boldsymbol{x}\|_1 \le n\|\boldsymbol{x}\|_\infty.
\end{aligned}
$$

   Find non-zero vector(s) to show that these inequalities are tight.

3. Consider the matrices $A \in \mathbb{R}^{5\times 4}$ and $B \in \mathbb{R}^{4\times 4}$ given by

$$
A = \begin{bmatrix} 3 & -2 & 0 & 5 \\ 1 & 8 & 7 & 0 \\ -9 & 5 & 1 & 2 \\ 3 & -2 & 0 & 5 \\ 3 & -4 & 0 & 5 \end{bmatrix}, \qquad B = \begin{bmatrix} 3 & -2 & 0 & 5 \\ 1 & 8 & 7 & 0 \\ -9 & 5 & 1 & 2 \\ 3 & -2 & 0 & 5 \end{bmatrix}
$$

   (a) For the matrix A calculate

      i. $\|A\|_1$ by hand and using Python.

      ii. $\|A\|_\infty$ by hand and using Python.

      iii. $\|A\|_2$ using Python.

   (b) For the matrix B

      i. How can it be efficiently calculated from the matrix A?

      ii. Calculate $\|B\|_1$, $\|B\|_\infty$, $\|B\|_2$

      iii. Calculate the largest magnitude eigenvalue $|\lambda_{\max}|$ of B.
      Does $\|B\|_2$ equal $|\lambda_{\max}|$?

   You may need to import the followings:

```
from numpy import inf, array
from scipy.linalg import norm, eigvals
```

4. For the following matrices, calculate the condition numbers with respect to the 1, 2, and infinity norms:

   (a) $\boldsymbol{A} = \alpha \boldsymbol{I}$ where $\alpha \in \mathbb{R}$, $\alpha \neq 0$ and I is the n by n identity matrix.

   (b) The $n \times n$ Hilbert matrix $\boldsymbol{H} = [h_{ij}]$ with entries

   $$h_{ij} = \frac{1}{i+j-1} \qquad i, j = 1, \ldots, n, \quad \text{in the cases} \quad n = 5, 10, 15.$$

   You may need to import the followings:

   from numpy import inf, identity, linalg

   from scipy.linalg import hilbert

5. Consider solving a linear system $\mathbf{A}\boldsymbol{x} = \boldsymbol{b}$ where the matrix A is known "exactly".

   (a) Estimate the number of significant figures in the computed solution when the right-hand-side vector $\boldsymbol{b}$ is known to 8 significant figures and

      i. $\|A\| = 1.2 \times 10^1, \quad \|A^{-1}\| = 3.9 \times 10^3$.

      ii. A is symmetric and has ordered eigenvalues $0.00031,\ 0.0045, \ldots,\ 101.4,\ 198.2$.

      iii. The reciprocal condition number of A is $9.9010e^{-12}$.

   (b) In each case, what must the relative error in $\boldsymbol{b}$ be to get a computed solution with 6 significant figures?

   You may need to import the followings:

   from numpy import finfo, log10, floor

6. Write a Python function symchk(A, tol) in a symchkSol.py file to determine if a matrix $\mathbf{A} = [a_{ij}] \in \mathbb{R}^{m \times n}$ is symmetric, that is, if $\mathbf{A} = \mathbf{A}^T$. Start the symchkSol.py file with:

   from numpy import maximum, any, finfo, float32, float64, ndarray

   def symchk(A, tol=None):

   Note the following.

   - The function returns True when the input matrix is symmetric to within a relative tolerance tol in the sense that $|a_{ij} - a_{ji}| \leq tol \max\{|a_{ij}, |a_{ji}|\}$ for all i and j, and False otherwise.
   - If the matrix is not square, then it cannot be symmetric.
   - If the input argument tol is not specified, use the default value $10n^2\epsilon$ where ϵ is the machine epsilon.
   - If the input argument tol is negative, raise a ValueError.
   - Create a Python file to test your function `symchk` on a variety of inputs.

   Here, you may need to import the following:

   from symchkSol import symchk

   from numpy import finfo, array, inf

   from numpy.random import rand, randn

   from scipy.linalg import norm